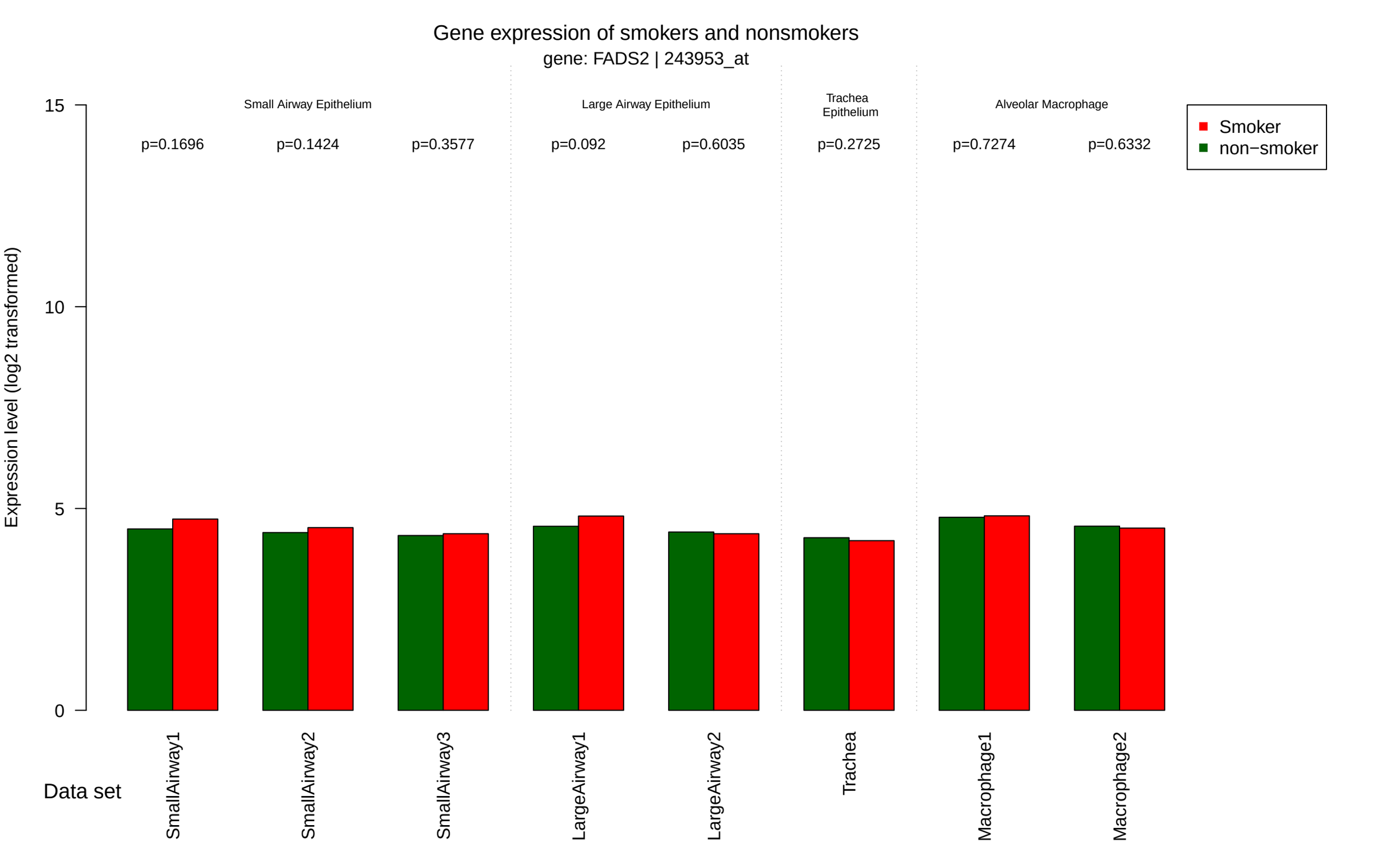
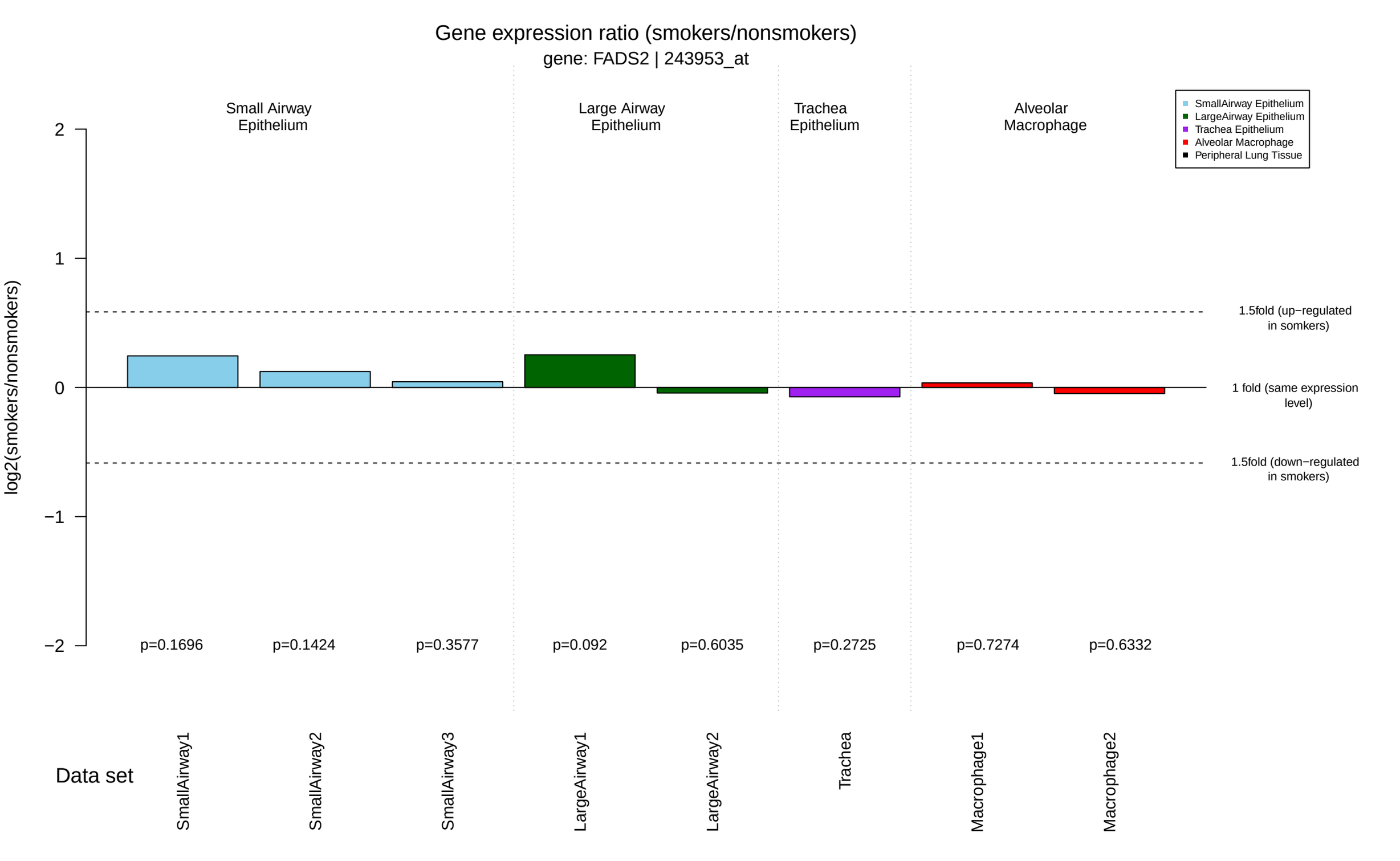
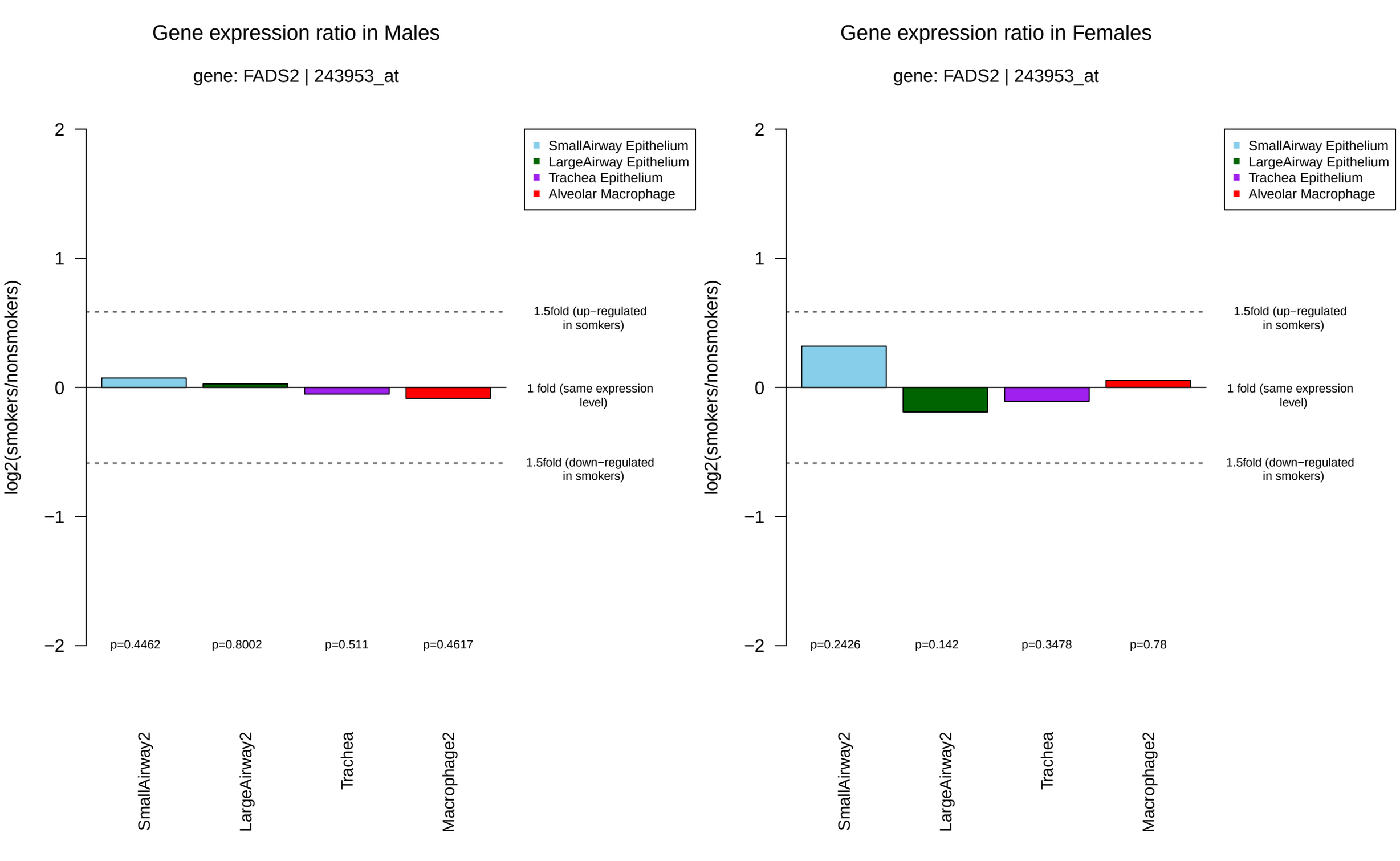
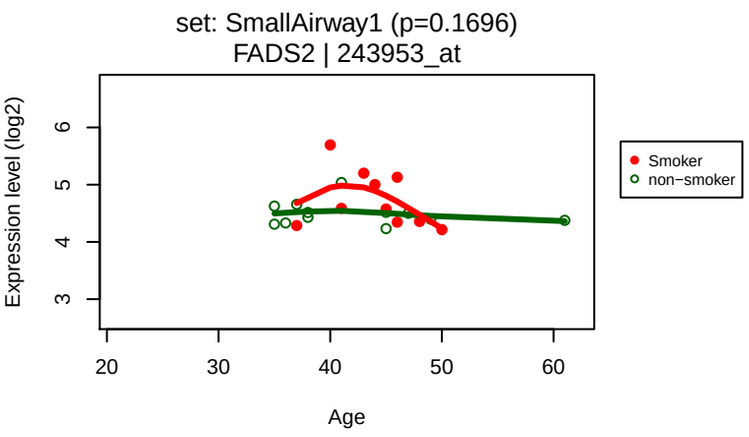
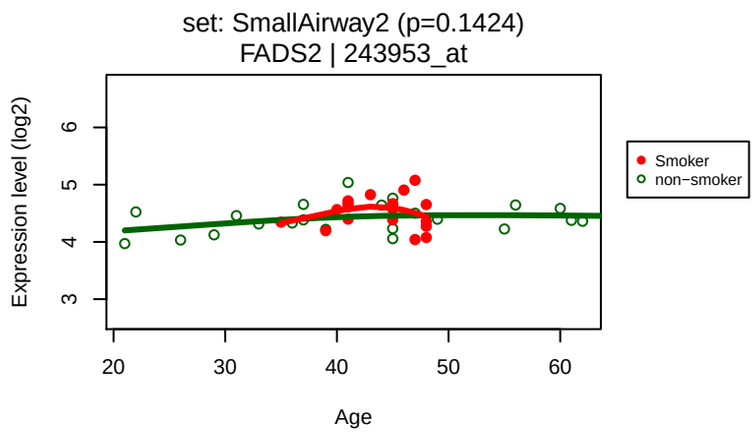
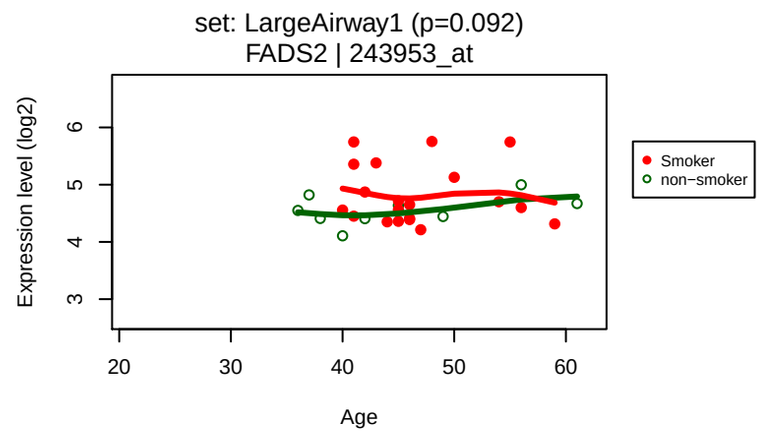

Gene expression of smokers and nonsmokers
gene: FADS2 | 243953_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1696
p=0.1424
p=0.3577
p=0.092
p=0.6035
p=0.2725
p=0.7274
p=0.6332
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

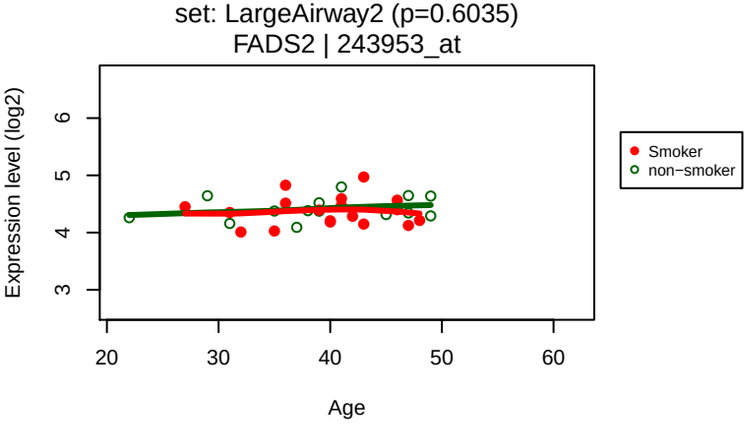

Gene expression ratio (smokers/nonsmokers)
gene: FADS2 | 243953_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1696
p=0.1424
p=0.3577
p=0.092
p=0.6035
p=0.2725
p=0.7274
p=0.6332
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

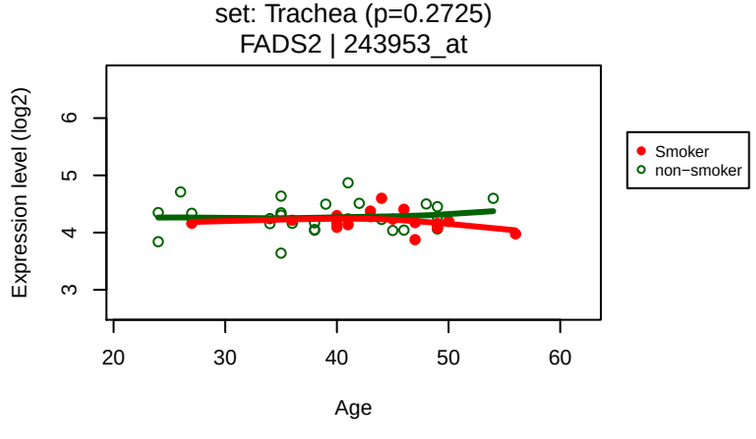

Gene expression ratio in Males
gene: FADS2 | 243953_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4462
p=0.8002
p=0.511
p=0.4617
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: FADS2 | 243953_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2426
p=0.142
p=0.3478
p=0.78
SmallAirway2
LargeAirway2
Trachea
Macrophage2

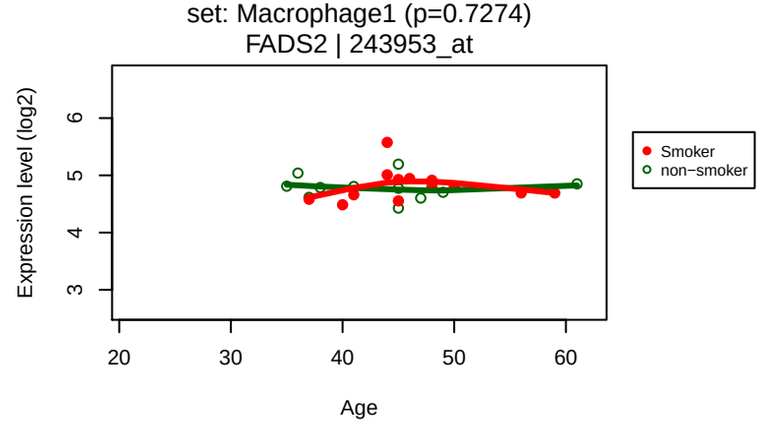

set: SmallAirway1 (p=0.1696)
FADS2 | 243953_at
Expression level (log2)
Age
Smoker
non−smoker

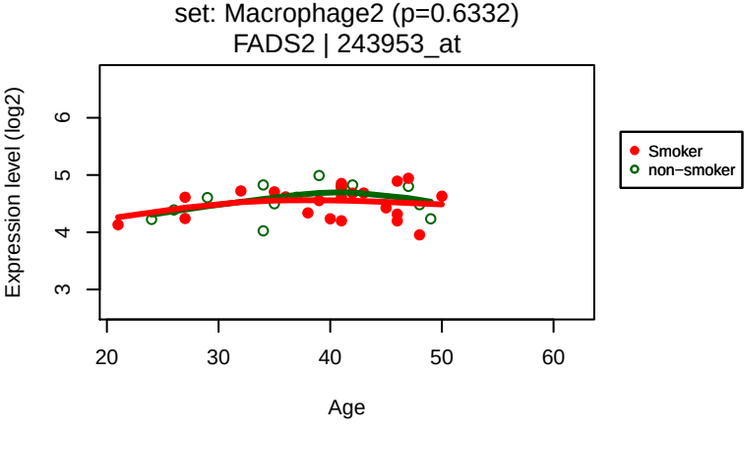

set: SmallAirway2 (p=0.1424)
FADS2 | 243953_at
Expression level (log2)
Age
Smoker
non−smoker

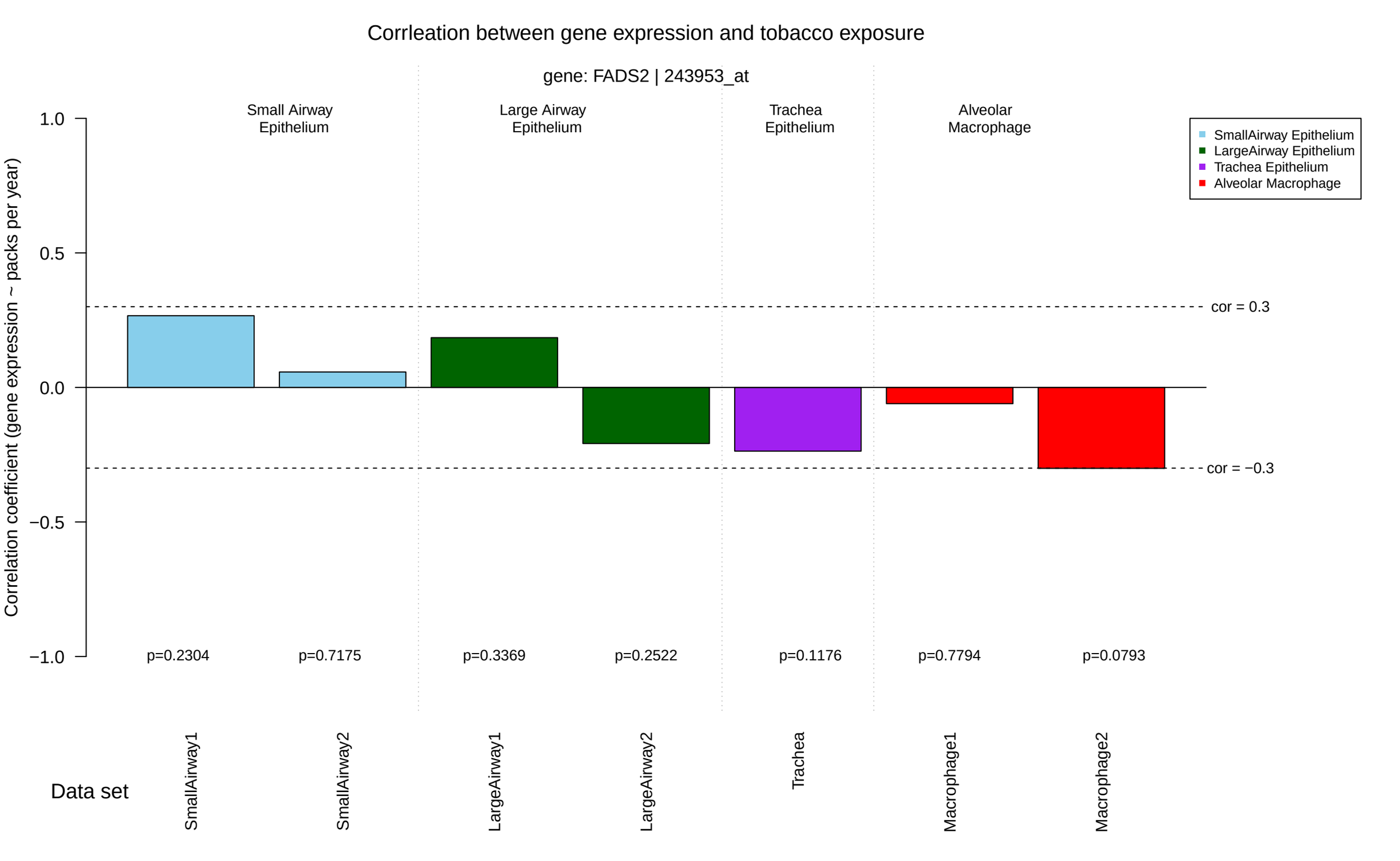

set: LargeAirway1 (p=0.092)
FADS2 | 243953_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6035)
FADS2 | 243953_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2725)
FADS2 | 243953_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7274)
FADS2 | 243953_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6332)
FADS2 | 243953_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: FADS2 | 243953_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2304
p=0.7175
p=0.3369
p=0.2522
p=0.1176
p=0.7794
p=0.0793
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2